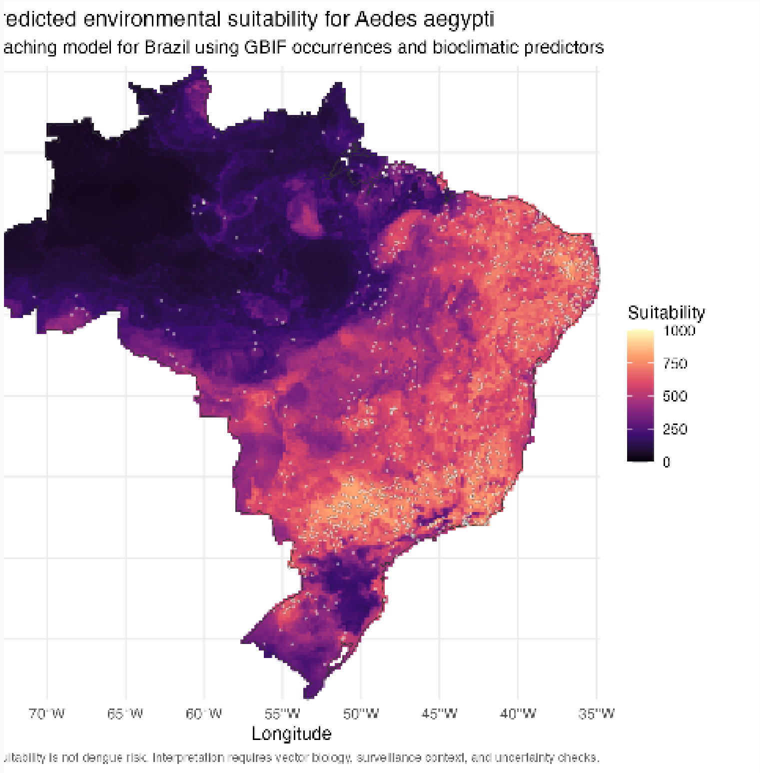
Mapping dengue vector suitability in Brazil

A beginner-friendly species distribution modeling workshop in R

Aedes aegypti + GBIF + WorldClim + biomod2



The public health question starts simple

Where could the environment support the vector?

For dengue, vector ecology is one layer of transmission risk. SDMs help us turn scattered observations into a map of environmental suitability.

Presence records

Where the mosquito has been observed

Environmental predictors

Climate layers across the study area

Suitability map

A hypothesis about where conditions fit

Today’s model follows one clear path

The workshop scripts keep each step visible.

1. **Download** GBIF records
2. **Clean** coordinates
3. **Load** predictors
4. **Fit** biomod2 model
5. **Map** suitability
6. **Discuss** limits

The important habit: inspect before you model, and interpret before you act.

GBIF gives us a starting point, not a perfect truth

Open biodiversity data are powerful, but they carry collection bias.

- **Useful**

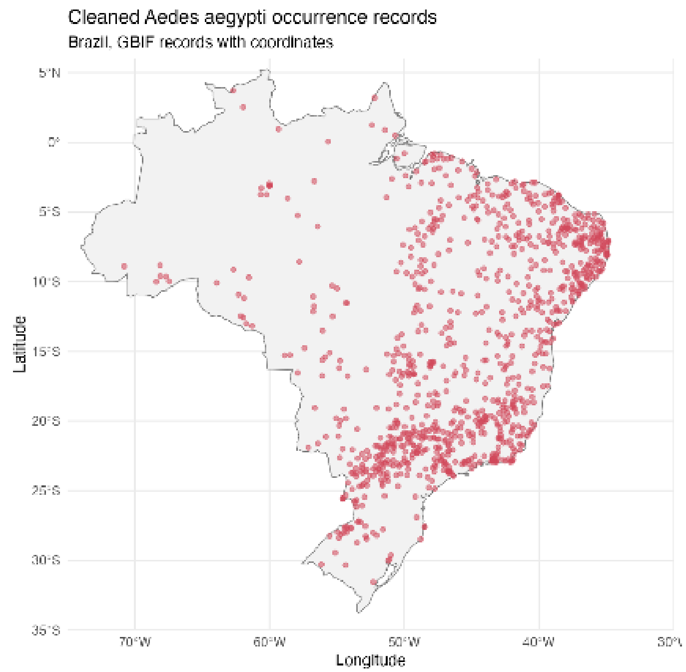
  Thousands of public records can be queried from R.

- **Messy**

  Coordinates may be duplicated, incomplete, or biased toward sampled places.

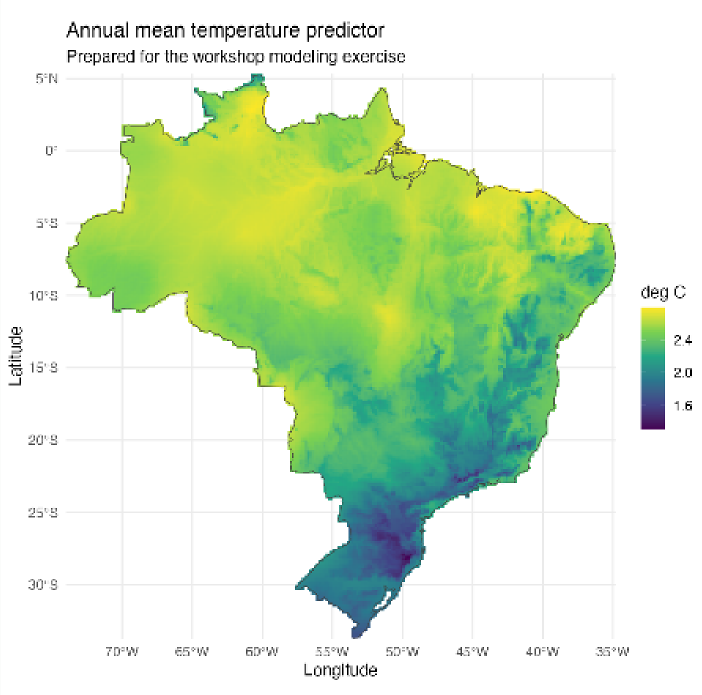
- **Teachable**

  Cleaning is part of the model, not a boring prelude.



Predictors describe the stage where the vector could live

We use a small set of climate variables so the first model stays readable.



Annual mean temperature

Temperature seasonality

Annual precipitation

Precipitation seasonality

Simple does not mean final. It means we can see what the workflow is doing.

biomod2 handles the modeling mechanics

Our job is to understand the assumptions we feed into it.

Format	presence points + environmental raster layers
Create	pseudo-absences for comparison
Fit	GLM, GBM, and RF examples
Evaluate	AUC and TSS as quick diagnostics
Project	suitability across Brazil

Live models

GLM

GBM

RF

+ ensemble

Pseudo-absences are a choice, not a fact

Presence-only data need comparison points, and that choice shapes the model.

What we know

These places have recorded Aedes aegypti.

They are presences, not a complete survey of Brazil.

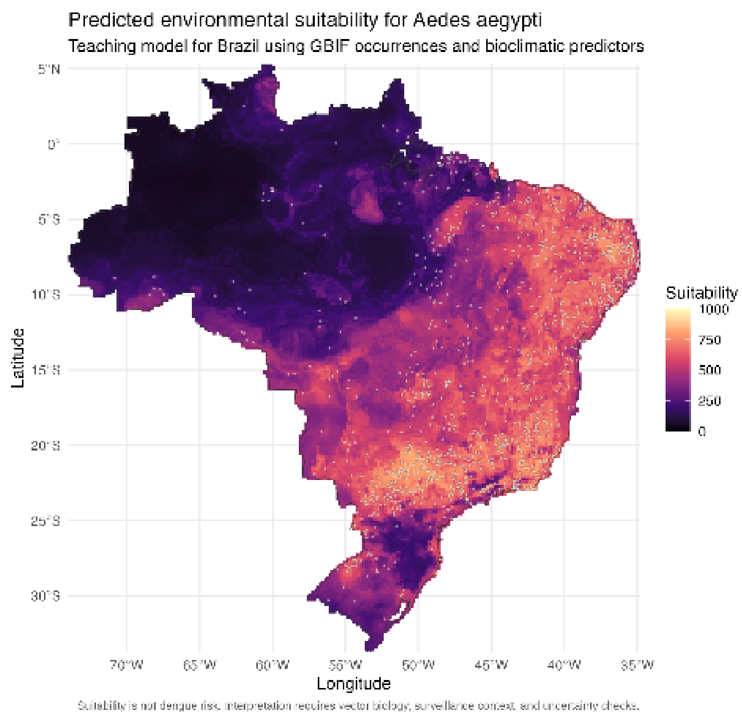
What we compare

Pseudo-absences describe available environments.

Different strategies can change the map.

This is why SDMs are scientific arguments, not map-making buttons.

Read the suitability map as a hypothesis



environments.

- Where is suitability high?
- Where are occurrence records sparse?
- What could be climate signal?
- What could be sampling bias?

A good SDM result should start a better conversation.

A vector SDM answers only part of the dengue question

That is a strength when we name the boundary clearly.

It can help with

- Vector suitability
- Surveillance prioritization
- Field sampling hypotheses
- Environmental comparison

It does not replace

- Case surveillance
- Vector abundance data
- Local control knowledge
- Transmission modeling

Your job today is not to become an SDM expert

It is to leave with a working map, a reusable R workflow, and better questions.

Run
the scripts line by line

Look
at the data before modeling

Ask
what the map can and cannot say

Let’s build the model.